

SWE 363 — Web Engineering · KFUPM

Portfolio Web Application

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alkerm.github.io/202279480-AbdulazizMahmoud-assignment4

Project Overview & Motivation



PROJECT OVERVIEW & OBJECTIVES

- ▶ Developed a personal portfolio web application
- ▶ Showcase projects, skills, and experience
- ▶ Provide an interactive and modern user experience



PERSONAL MOTIVATION

- ▶ Needed a professional online presence
- ▶ Wanted to demonstrate real-world development skills
- ▶ Opportunity to integrate modern web technologies and AI features

Challenges, Solutions & Lessons

CHALLENGES & FIXES

- **Responsive UI:** min-width: 0 on grid cells prevented card overflow on mobile
- **Smooth animations:** max-height + opacity transitions replaced display:none
- **API timeouts:** AbortController with 8s timeout + error fallback UI
- **Filter/sort wiring:** Unified inputs to call applyFiltersAndSort; reflow trick restarts animations

UNRESOLVED & FUTURE

- **Backend :** complete some functionalities by implementing the backend.
- **Micro-animations:** Staggered entrances + parallax via IntersectionObserver
- **More Sections:** adding more sections that would add value to the portfolio.

LESSONS LEARNED

- **Plan first:** Sketching component trees prevented most rewrites
- **Clean prompts:** Scoped AI prompts gave usable output; vague ones didn't
- **DevTools always:** Network tab + Lighthouse caught what code review missed

Outcomes, Achievements & Next Steps



Live Deployment

Publicly accessible portfolio at a GitHub Pages URL



7 Sections Built

Hero · Timeline · Projects · Skills · GitHub Repos · UCL Widget · Contact



Fully Responsive

Tested across desktop, tablet, and mobile — down to 320 px



Professional Polish

Custom favicon, Open Graph meta tags, GPU-optimised sticky header, theme toggle



Creative Innovation

Animated timeline, scroll-shrinking header, real-time API widgets — zero hardcoded data